

WP3: Filtering and Clustering of AI-Generated Enzyme Mimics

- Modern AI-based enzyme design pipelines can generate thousands of enzyme candidates.
- Experimental validation is expensive and time-consuming.
- Most generated structures exhibit limited structural diversity.
- **Goal of WP3:**
 - Preserve catalytic functionality;
 - Maximize structural diversity;
 - Select only the most promising candidates for laboratory validation.

Position of WP3 within the Workflow



Generate → Sequence Design → Structure Prediction → Candidate Selection → Experimental Validation

Main Objective

Explore the structural space of enzyme mimics while preserving catalytic functionality.

Two complementary strategies are employed:

1 Filtering

- Improve candidate quality.
- Remove unlikely functional structures.

2 Clustering

- Promote structural diversity.
- Ensure broad exploration of enzyme structural space.

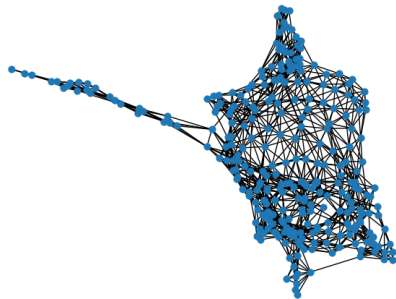
Protein Structure as a Graph

- Each amino-acid residue becomes a graph vertex.
- Residues are connected if their C_α atoms are spatially close.
- Threshold:

$$\tau = 8\text{\AA} \text{ (ångström unit)}$$

- Produces a residue interaction graph.

$$(v_i, v_j) \in E \iff \|x_i - x_j\|_2 \leq 8\text{\AA}$$



Graph-Based Structural Features

Basic Features

- Number of nodes
- Number of edges
- Density
- Average degree
- Minimum degree
- Maximum degree

Connectivity Features

- Connected components
- Diameter
- Average shortest path

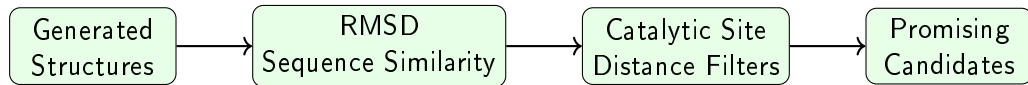
Topological Features

- Clustering coefficient
- Degree centrality

Advanced Features

- Catalytic residue geometry
- Minimum Dominating Set size
- Protein language model embeddings
- Structural descriptors

Filtering Pipeline



- Remove structurally implausible candidates.
- Preserve catalytic residue organization.
- Focus computational resources on promising structures.

Catalytic triad of turboPETase

SER – HIS – ASP

Filter 1

- Compare catalytic residue positions between generated and reference enzymes.

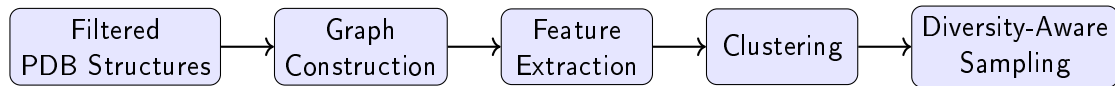
Filter 2

- Measure internal distances:
 - SER–HIS
 - HIS–ASP

Objective

- Preserve catalytic geometry.
- Increase likelihood of catalytic functionality.

Clustering Pipeline



- 1 Convert structures into graphs.
- 2 Compute structural descriptors.
- 3 Build feature space.
- 4 Identify structural families.
- 5 Select representatives from each cluster.

Clustering Methods

- K-Means
- Hierarchical Clustering
- Spectral Clustering
- Density-Based Clustering

Cluster quality metrics

- Silhouette score
- Intra-cluster similarity
- Inter-cluster separation

Expected Outcome

- Distinct structural families of enzyme mimics.

Feature Space Representation

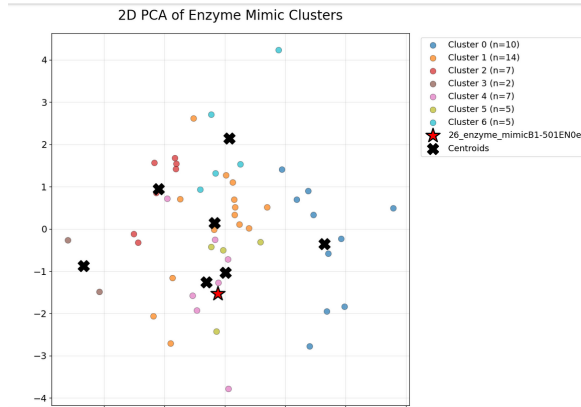
0	6_enzyme_mimicB1-501EN379c	308	1591	0.033652	10.331169	4	19	0.533014	1	0.033652
1	11_enzyme_mimicB1-501EN53d	324	1632	0.031189	10.074074	4	18	0.547730	1	0.031189
2	14_enzyme_mimicB1-501EN28d	335	1659	0.029654	9.904478	4	19	0.544153	1	0.029654
3	8_enzyme_mimicB1-501EN53a	325	1599	0.030370	9.840000	2	19	0.549844	1	0.030370
4	42_enzyme_mimicB1-501EN349a	319	1607	0.031683	10.075235	3	19	0.537129	1	0.031683
5	28_enzyme_mimicB1-501EN60a	324	1613	0.030826	9.956790	2	19	0.544257	1	0.030826
6	34_enzyme_mimicB1-501EN59e	312	1540	0.031742	9.871795	2	19	0.555165	1	0.031742
7	17_enzyme_mimicB1-501EN59a	313	1579	0.032338	10.089457	2	19	0.540277	1	0.032338
8	15_enzyme_mimicB1-501EN20e	334	1622	0.029167	9.712575	4	18	0.545836	1	0.029167
9	31_enzyme_mimicB1-501EN321e	328	1655	0.030861	10.091463	2	19	0.548219	1	0.030861
10	30_enzyme_mimicB1-501EN259d	324	1582	0.030234	9.765432	3	19	0.545546	1	0.030234
11	13_enzyme_mimicB1-501EN60c	324	1633	0.031208	10.080247	2	18	0.536101	1	0.031208
12	19_enzyme_mimicB1-501EN28b	334	1648	0.029634	9.868263	4	19	0.541038	1	0.029634
13	44_enzyme_mimicB1-501EN37d	330	1666	0.030690	10.096970	2	19	0.540706	1	0.030690

Protein structures represented through graph-based and biochemical descriptors.

K-Means Analysis

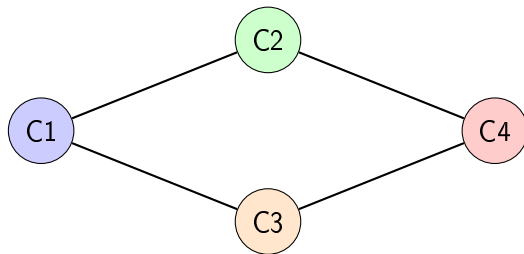
```
k = 2, silhouette = 0.2424
k = 3, silhouette = 0.2076
k = 4, silhouette = 0.1974
k = 5, silhouette = 0.2168
k = 6, silhouette = 0.2367
k = 7, silhouette = 0.2680
```

Silhouette analysis used to determine the number of clusters.



2D PCA visualization of clustered enzyme structures.

Consensus-Based Candidate Selection



- Select representative structures from each cluster.
- Avoid selecting many variants of the same fold.
- Increase structural diversity of experimentally tested candidates.

- Reduce wet-lab validation costs.
- Increase structural diversity of enzyme mimics.
- Preserve catalytic functionality.
- Introduce graph-theoretical and combinatorial descriptors into enzyme design.
- Enable systematic exploration of protein structural space.
- Establish a feedback loop between AI generation and experimental validation.

Towards Reliable and Diverse AI-Generated Enzymes